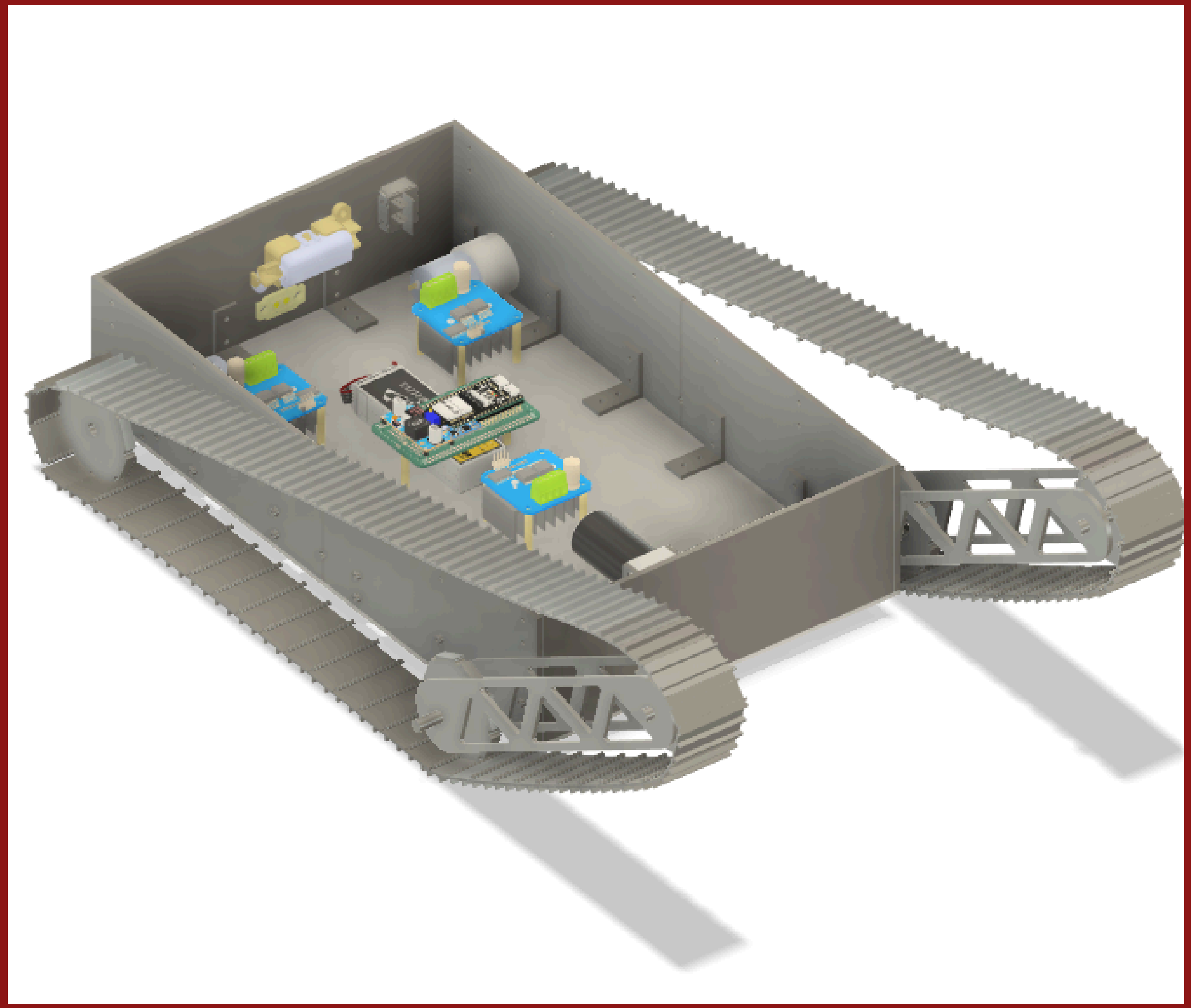
# Stair Climbing Autonomous Robot Delivery (SCARD)



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# Purpose of SCARD

### Overcoming Delivery Challenges in Dorms

Dorm deliveries are often **inefficient** due to stairs and limited access. Shot Bot addresses these obstacles, providing a streamlined solution for timely and effective deliveries within campus environments.

# Core Features

**Key functionalities of SCARD**

### Autonomous Navigation

Shot Bot utilizes advanced sensors and algorithms to navigate autonomously, allowing it to efficiently traverse complex environments within dorms without human intervention.

### Stair Climbing

Engineered with innovative flipper-track mechanics, SCARD can conquer stairs, ensuring smooth delivery regardless of elevation changes encountered in dormitory buildings.

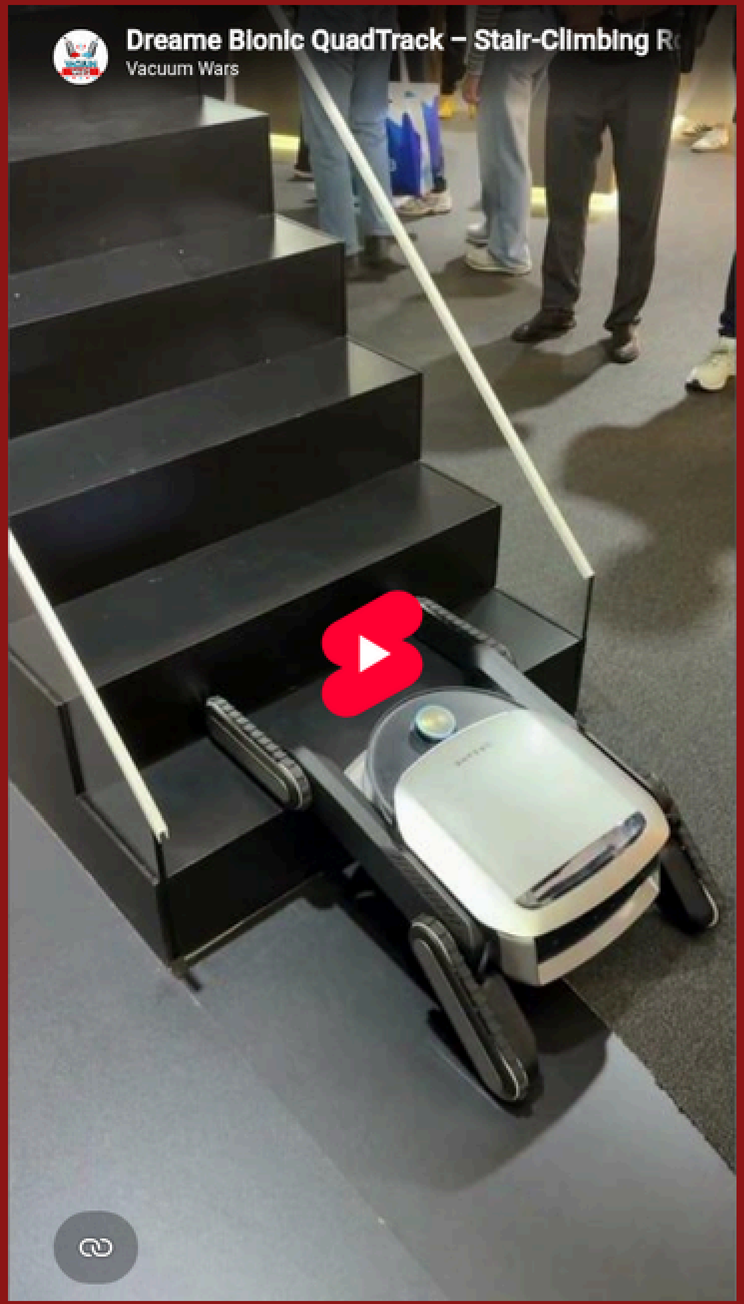
### Tracked Motion

The robot's tracked motion system provides stability and traction, enabling it to maneuver over various surfaces while maintaining balance and avoiding potential obstacles.

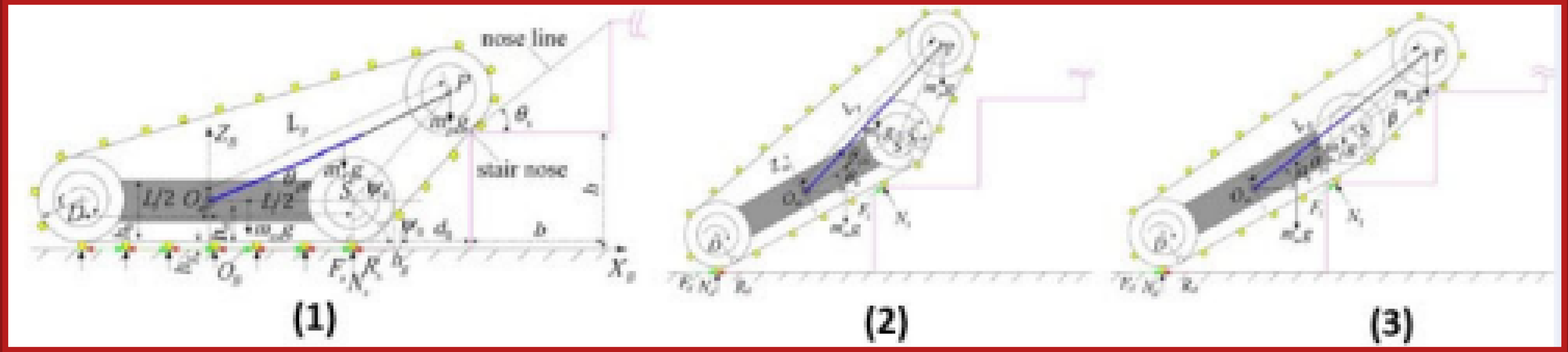
# Mechanical System Overview

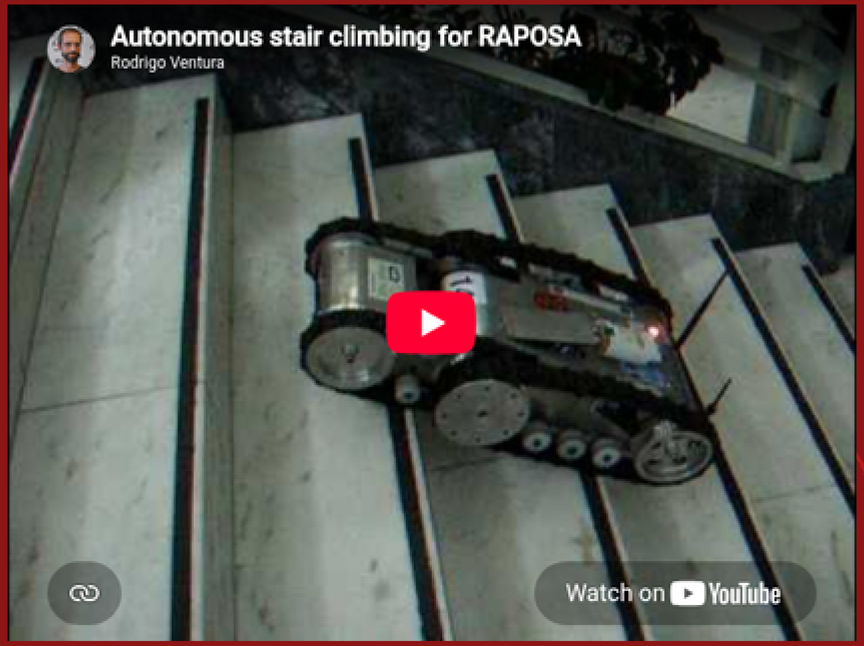
### Understanding SCARD Movement Mechanics

The SCARD employs a **unique design** using flipper arms, enabling efficient stair navigation with a sprocket layout that maintains track integrity.





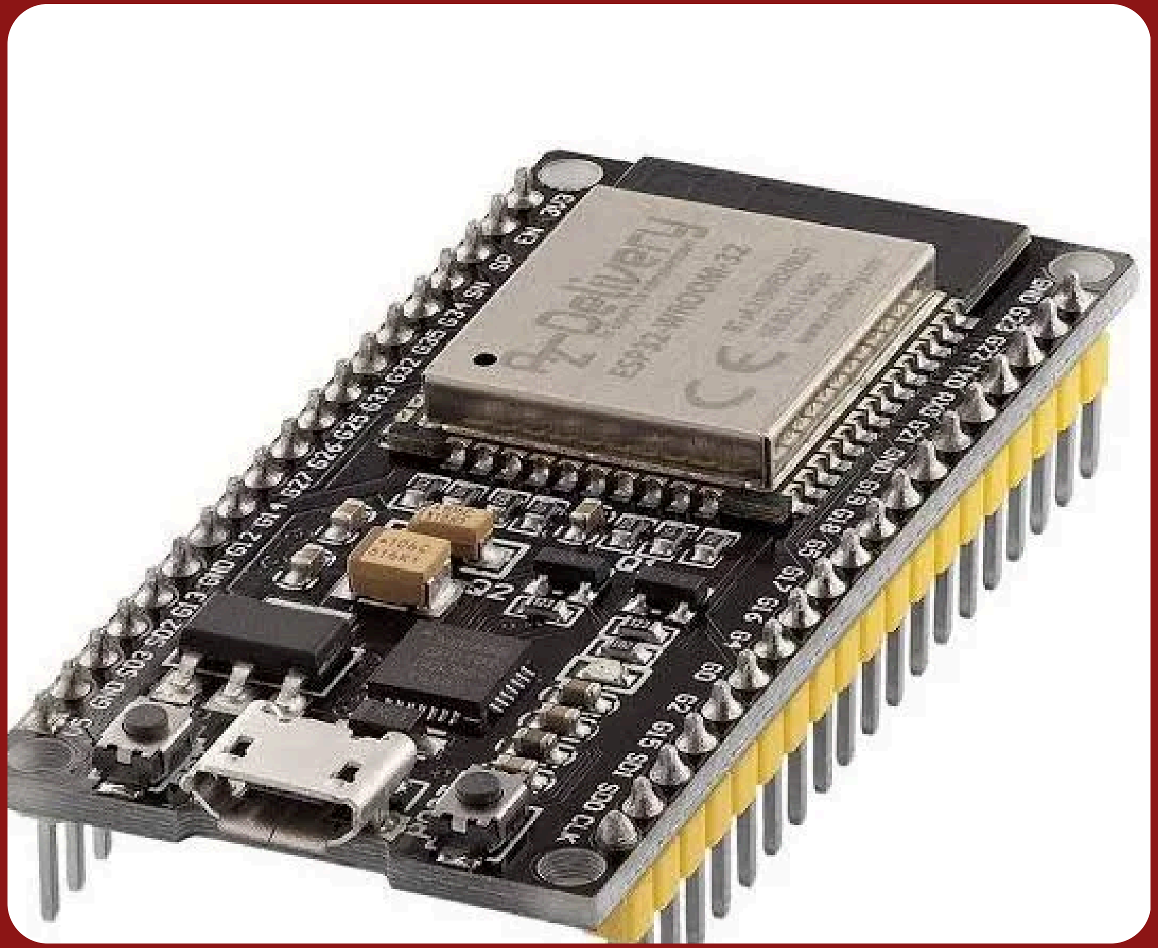


# Electronics Control Systems

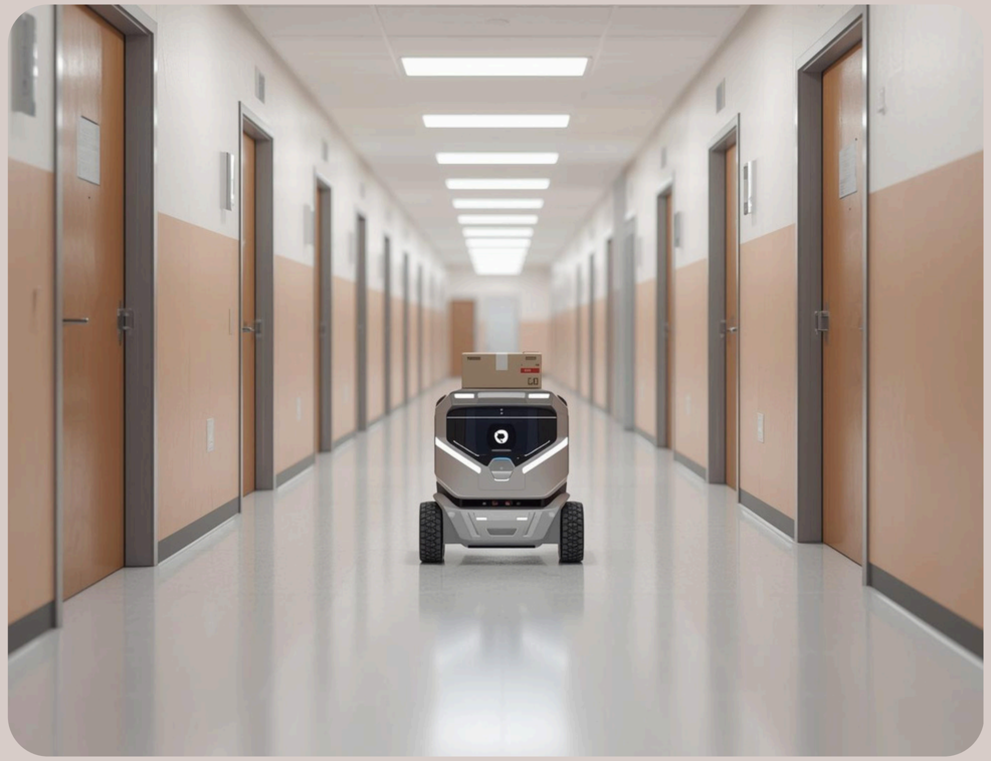

## Integrating Smart Technology for Navigation

SCARD utilizes an **ESP32 microcontroller** to manage motor drivers and sensors, including LiDAR and camera, ensuring efficient navigation and delivery within dorm environments.



# Applications of Shot Bot

## Dorm Delivery



SCARD streamlines **dorm deliveries**, ensuring timely and efficient package drop-offs for students.

## Lab Supply Transport

Used for **lab supply transport**, SCARD efficiently manages deliveries, enhancing research workflows and productivity.

## Robotics Research

SCARD serves as a tool in **robotics research**, contributing to advancements in autonomous navigation technology.

# First Prototype Goals

- Stair Climbing Capability: Designed to effortlessly navigate staircases for use in multi-level settings.
- Remote Operability: Controlled remotely for convenient manipulation and direction.
- Item Transport: Capable of carrying and moving various objects efficiently.